

Overview of the Policy and Its Historical Context

As part of its 2025–2030 'Forward Together' strategic plan, UNCG has committed to enhancing student well-being, fostering 'best-in-class undergraduate degree attainment for students from all backgrounds,' and preparing students for meaningful careers and lives (UNCG, 2025, pg. 7–8). The university also aims to increase four-year graduation rates, achieve R1 Carnegie classification, and elevate its brand and reputation (UNCG, 2025, pg. 4, 7, 12). Fulfilling this vision will require innovative, community-engaged pedagogical approaches – approaches that empower students as agents in their own success while addressing both their fundamental needs and those of the local community. The growing emphasis on 'Civic Learning' and 'Community Engagement' in higher education reinforces this priority, as reflected in the expanded focus on these areas in the 2026 Carnegie Classification application (Carnegie Foundation, 2024, pg. 14–17). Achieving these goals has the potential to generate significant economic benefits for both the university and its students, enhancing the university's reputation, reducing financial barriers to student success, and fostering long-term economic mobility. This paper explores the theoretical foundations and frameworks that support piloting a community-engaged, student-run Free Store as a theoretical model for advancing these outcomes.

Free Stores, a concept dating back to the 1960s, aim to provide essential goods to the local community while simultaneously diverting usable goods away from landfills. Inventory is sourced through a wide range of channels, including food recovery programs, community donations, and corporate partnerships, and is presented in a traditional retail format. This model has the potential to not only reduce market inefficiencies created by waste and lack of access, but also return agency to customers, positioning them as consumers rather than recipients of aid. Furthermore, by engaging students in the design, operation, and evaluation of this resource, the Free Store can evolve into a peer-driven, innovative, and self-evaluatory learning lab. Such a project offers long-term benefits, not only for the students who organize and patronize the store but also for institutions seeking to augment their community engagement efforts while strengthening their impact metrics.

The interdisciplinary nature of this project necessitates research from multiple fields, each offering insights that coalesce into a framework for reducing barriers to student success and supporting the university's role in community-engaged education. The following sections integrate literature, modeling, and evaluation to examine the viability of piloting a campus-based Free Store.

Reviewing the Literature and Its Broader Connections

The existing literature on student support programs, mutual aid models, and campus engagement provides a foundation for evaluating how a Free Store might influence student retention and enrollment yield. While there are no studies that specifically focus on campus-based Free Stores, research across multiple fields helps to clarify the mechanisms involved and the outcomes that can be reasonably anticipated.

Access, Stigma, and Structural Barriers

Idehai et al. (2024) conducted a systematic review of eight studies that examine predictors, barriers to, and facilitators of campus food pantry use. Their analysis found that students who are food insecure, receive financial aid, are first-generation, or identify as international or non-white are significantly more likely to use pantry resources. However, the authors also emphasize that structural and individual forms of stigma, limited access to culturally appropriate food, and logistical constraints (such as lack of kitchen infrastructure) present meaningful barriers. Messaging from institutions themselves, often rooted in competitive or meritocratic narratives, can inadvertently reinforce stigma. The study concludes that addressing stigma and improving access to quality, culturally relevant food are key priorities for improving utilization and outcomes.

Free Stores and Identity Repair

In a case study conducted in New Zealand, Dresler and Tutt (2019) explore how participation in Free Store operations fosters identity development and reduces stigma. The study focuses on what they call the “customer to volunteer transition:” a recurring theme among participants who reported feeling a desire to give back after receiving aid. The authors highlight how the store’s no-means-testing policy and community-oriented structure encouraged both participation and belonging. Although not focused specifically on college students, the findings provide valuable insight into how exchange-based resource models can offer a more empowering experience than traditional aid programs.

Institutional Priorities and Public Engagement

UNCG’s *Forward Together* strategic plan lays out goals tied to retention, graduation, and public recognition, with a particular emphasis on equity, student well-being, and civic learning. The plan’s emphasis on becoming a national model for undergraduate

success and increasing institutional prestige highlights the need for low-cost, high-impact programs that demonstrate support for diverse student populations. These goals are echoed in the updated Carnegie Community Engagement Classification application, which foregrounds the role of campus-community partnerships in enhancing educational outcomes. A Free Store operated as a Civic Learning Lab, where students actively participate in forwarding its mission, would not only contribute to these goals but also expand opportunities for community-engaged research, service learning, and experiential education.

Participatory Design and Curriculum Integration

Both Barr and McNamara (2022) and Coles-Ritchie et al. (2022) explore how student participation in the design and implementation of campus programs can improve buy-in, relevance, and learning outcomes. Using participatory and community-based frameworks, these studies demonstrate that students who co-create programming are more likely to view it as meaningful and are better equipped to sustain it. These findings directly support the idea that a Free Store should not be a top-down service, but rather a peer-driven space shaped by evolving student needs. These papers also show how structured reflection and curriculum integration can turn a service space into a site of interdisciplinary learning and student development.

Evaluation and Measurement

One recurring theme across the literature is the need for better measurement tools to evaluate civic learning and well-being outcomes. Janke and Medlin (2015) document the development of UNCG's campus-wide engagement data infrastructure, focusing on how to track public service efforts across faculty and staff. Benjamin et al. (2024) take a broader view, discussing the limits and opportunities of using self-reported happiness data in economic research. Both papers speak to the importance of thoughtful, well-designed data collection, especially when measuring outcomes that may be qualitative or long-term in nature. These insights are particularly valuable for assessing a Free Store's contribution to student belonging, pride, and retention.

Conclusion

Together, these studies suggest that destigmatized, community-oriented resource models that actively involve students in design and implementation may lead to higher participation, stronger identity formation, and more equitable outcomes. Although additional research is needed to test these ideas in the context of a campus Free Store,

the existing literature provides a strong foundation for predicting positive impacts on both student engagement and institutional success.

Constructing a Model to Evaluate Institutional and Student Outcomes

Agents

University

Institutional Leadership – Decision-makers on campus assess potential programs, ensuring that proposals align with institutional priorities before approving or rejecting their implementation. In the context of a proposed Free Store, the institution would focus on cost-benefit analyses, implementation strategies, impact on institutional identity, and ultimately, whether to establish the free store in the first place.

Administration – The modeled costs include a full-time salaried position (paid at \$65k annually) responsible for synthesizing instructor, student, community, and institutional needs – aiming to maximize overall benefits. Responsibilities include facilitation of student discussions, co-development of projects for various disciplines, engaging community partners, and maintaining a database of performance metrics. The Free Store will also need regular cleaning, insurance, and assistance in tracking necessary data. These tasks engage a variety of university staff yet are not particularly atypical and should integrate well with existing workflows.

Instructors – The engaged learning component of academic internships requires a collaborative relationship between the student, site leader (manager), and instructor. Student placement, ongoing communication, and tracking of learning outcomes are difficult to control, creating a logistical burden. However, the Free Store offers a variety of experiential learning opportunities, with the added benefit of in-house communication channels and guidelines. In theory, this reduces the workload for instructors who already facilitate external internship placements.

Students

Though the distinctions between students who run the Free Store and students who patronize it are minimized in practice due to the operational structure, it is still worthwhile (especially for initial users) to distinguish their actions and motivations

Patrons – Students who patronize the Free Store can be motivated by traditional factors such as food insecurity and economic hardship but also include those who are interested in peer-driven resources, especially those that better reflect their needs as a result. Additionally, students who desire alternatives to disposal of unneeded items can donate those goods, ensuring the remaining useful life does not go to waste.

Interns – Student interns engage with the Free Store for academic credit, interest in community-engaged learning, an opportunity to participate in the development of their curriculum, or to contribute to academic discourse surrounding its operations. For students without pre-existing professional networks, the Free Store internship provides an opportunity to develop practical, marketable social and civic skillsets.

Parameters

Benefits:

Retail value of items sourced (monies not spent by students) – These values diminish as usage increases. At .05% utilization, those with the highest need are likely to be patrons of the resource. This model assumes \$150 of retail value (V) obtained per visit. At higher levels of utilization, diminishing student need and limitations in storefront capacity combine to reduce per visit value. Each student is expected to visit the store twice monthly during the Spring and Fall semesters (16 total visits annually).

Retained Tuition – Through cross-sectional analyses, Kane (2003) identified a range of 3% – 5% increased enrollment for every \$1000 (\$2001) in financial aid and Nguyen et al. found a 1.5% – 2% increase in retention rates per \$1000 (\$2019). To account for these benefits, it is assumed that the retail cost of items received for free serves as a proxy for financial aid. In the baseline model, these benefits are calculated by adjusting the aid values for inflation, utilizing the lowest percentages in ranges for both enrollment and retention (3% and 1.5% respectively), and assuming only two additional years of tuition paid at the in-state level.

Variable Definitions:

S = Number of participating students

V = Total annual retail value of sourced items

E = Enrollment increase from financial aid (assumed 3%)

C = Completion increase from financial aid (assumed 1.5%)

T = Tuition per semester (\$7731)

Y = Number of years retained (assumed 2)

R = Retained students (rounded down)

TU = Retained tuition revenue (\$)

Step-by-Step Formulas

Enrollment Increase per \$1000 (\$2001) in aid (per \$1814.23 adjusted for inflation)

$$E = (S \times 0.03) \times (V \div 1814.23)$$

Completion Increase per \$1000 (\$2019) in aid (per \$1262.04 adjusted for inflation)

$$C = (S \times 0.015) \times (V \div 1262.04)$$

Total Retained Students (Rounded Down)

$$R = \lfloor E + C \rfloor$$

Retained Tuition Revenue Calculation

$$TU = R \times (T \times 2Y)$$

Combined Formula

This formula estimates the total tuition revenue (TU) generated by the Free Store resulting from increased student enrollment and retention. It combines two separate effects:

1. A 3% enrollment increase for every \$1,814.23 in goods provided (adjusted from Kane, 2001), and
2. A 1.5% retention increase for every \$1,262.04 in goods provided (adjusted from Nguyen et al., 2019)

The formula first calculates the number of additional students generated through these two effects, based on the number of participating students (S) and the total value of

goods received per student (V). These effects are then rounded down to the nearest whole student.

The resulting number of additional students is multiplied by tuition per semester (T) and the number of semesters retained (Y) to yield the total retained tuition revenue (TU) attributable to the Free Store.

$$TU(S,V,T,Y) = [(S \times 0.03) \times (V \div 1814.23)) + ((S \times 0.015) \times (V \div 1262.04))] \times (T \times Y)$$

Costs:

Staff Labor – Full time position at \$65k annually. Serves as site leader for student internships, networks with local community to seek donations, creates awareness campaigns. Their job is to maximize utilization.

Space utilization – Calculated at 800sq/ft at \$30 per sq/ft annually

Supplemental Inventory – Purchases made by institution to fill out storefront selection with necessary supplies (writing equipment, notebooks, branded winter wear, retail displays, etc.) calculated as 25% of the retail value of outgoing items. The 25% assumption is extremely high based on traditional Free Store models which operate exclusively on donation. This value is meant to capture the unique costs of ensuring the availability of essential school supplies while also providing for the purchase and maintenance of retail infrastructure including shelving, POS systems, and marketing materials.

Soft Outcomes

Due to methodological disagreements on valuation, psychological and social well-being benefits are omitted, though they likely have a substantial effect on student outcomes and institutional prestige. The relationships between these parameters and their expected effects are very briefly explored below.

Soft Benefits

Increased self-confidence – Gains in academic performance and workforce readiness outcomes

Reduction in stigma – Gains in self-confidence, increased utilization of assistance resources

Accessible Internship Opportunities – Increased chances of job placement and gains in earning potential

Institutional prestige and reputation gains – Increased enrollment, tuition revenue, graduate success, and grant funding

Soft Costs

Perceived Ideological Affiliation and Public Relations Risk – The Free Store's model is historically rooted in mutual-aid, anti-capitalist, and leftist movements. It is reasonable to assume that many community or national actors may fundamentally disagree with the political and economic ideologies of these movements. The best way to speak to these concerns is demonstrating profitability to the institution in real dollars, under conservative assumptions.

Campus Retailers – Students who source clothing or food from the Free Store are less likely to patronize campus-based retailers to meet those needs. At scale, a Free Store could create measurable decreases in sales at retailers that contract with the university, potentially opening the door to strained relationships with existing partnerships.

Marginal Analysis of Program Efficiency

The Free Store's economic impact is modeled across a range of student utilization levels, from 0.5 to 4.0 percent of the undergraduate population. This analysis centers on marginal benefit and marginal cost, providing insight into the program's efficiency at different levels of participation. Marginal benefit includes both the estimated retail value of goods received by students and tuition revenue generated through improvements in enrollment and retention. These gains are calculated using conservative estimates derived from Kane (2003) and Nguyen et al. (2019), which link increases in financial aid to gains in persistence and enrollment.

Marginal cost includes both fixed and variable expenditures. Fixed costs consist of one full-time staff position and 800 square feet of campus space allocated to the program. Variable costs scale with participation and assume a 25% supplementation of the retail value of goods distributed. While the benefits accrue to both students and the university, the model is designed to evaluate the institutional return on investment, recognizing the university's dual priorities of student support and financial sustainability.

At low levels of participation, marginal benefit significantly exceeds marginal cost. This result reflects both the presence of high-need users and the distribution of fixed costs across a smaller user base. As participation increases, marginal cost gradually declines due to economies of scale. However, marginal benefit also begins to fall. Students who participate later in the adoption curve are less likely to face acute material need, and the institutional return associated with their participation is correspondingly lower.

In addition, logistical constraints emerge as participation rises. The fixed size of the storefront limits both physical capacity and throughput, while sourcing and managing sufficient inventory becomes more complex at higher utilization levels. These structural frictions contribute to diminishing marginal benefit, even as aggregate program value continues to rise.

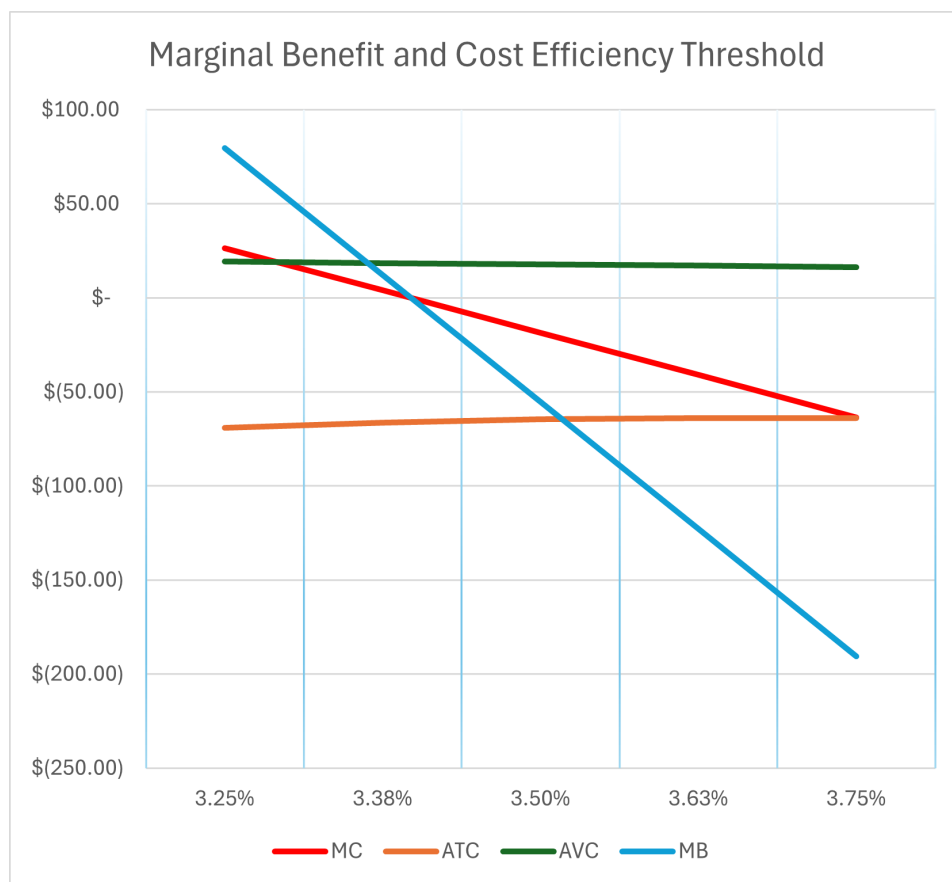


Figure 1 illustrates the range between 3.25% and 3.75% utilization, where marginal benefit intersects marginal cost. At approximately 3.38%, the program reaches its efficiency threshold. Beyond this point, additional participation continues to generate net institutional benefit in aggregate, but at decreasing marginal efficiency. The inflection point signals the limit of the current program design's ability to generate proportional returns and should inform any decisions to scale the model further.

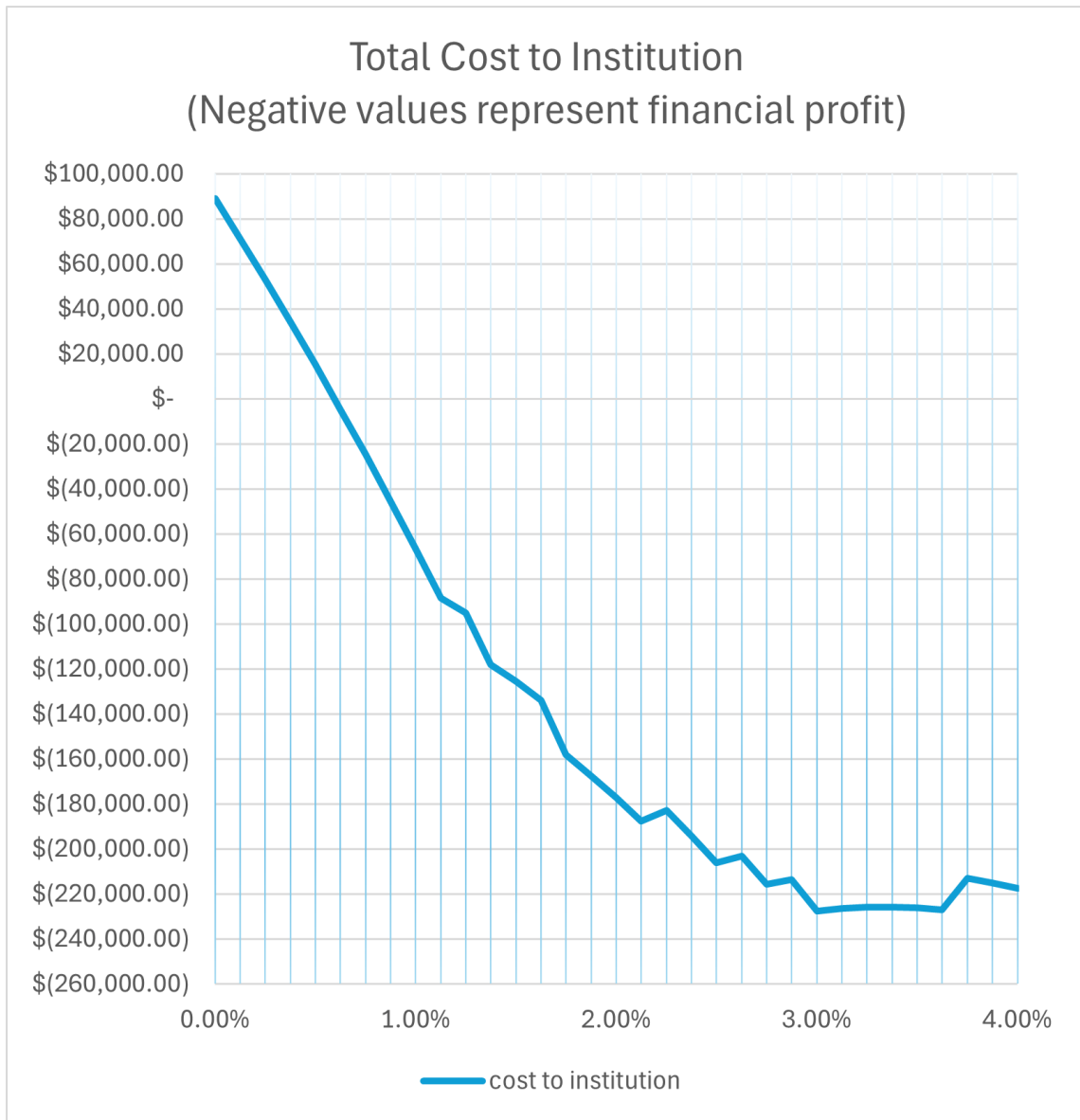


Figure 2 presents total institutional net cost or profit across the utilization range. The model shows the Free Store breaks even at approximately 0.68% utilization—roughly 120 students. Above that threshold, institutional returns rise steadily. These returns are calculated without incorporating gains related to student identity, internship access, or institutional reputation.

Data Sources for Testing the Model's Predictions

To test the hypothesis that access to a campus-based Free Store improves student retention and enrollment yield, this project draws on three distinct types of data: public institutional data, internal metrics, and survey data that would be collected as part of the Free Store's operations. Together, these sources allow for analysis of key student success outcomes, as well as insight into mechanisms like belonging and perceived support.

IPEDS (Integrated Postsecondary Education Data System)

IPEDS provides standardized, longitudinal data on institutional performance, allowing for the analysis of trends over time and comparisons across peer institutions. Two variables are particularly useful for testing the hypothesis:

DRVADM20xx_RV: Admissions yield – total

This measures the percentage of admitted students who ultimately enroll at UNCG. Yield is an important signal of institutional appeal, especially for marginalized students evaluating whether they will be supported.

EF20xxD_RV: Full-time retention rate

This measures the percentage of first-time, full-time undergraduates who re-enroll at UNCG the following year. Because IPEDS data is consistent and publicly available, it allows for institutional-level time-series analysis that could track the impact of a Free Store on retention and yield before and after implementation.

UNCG Retention, Graduation, and Persistence Dashboards

In addition to public datasets, UNCG's internal dashboards provide data on student success outcomes across key demographic categories. This includes retention, graduation, and persistence rates broken down by Pell eligibility, race/ethnicity, and other markers of student vulnerability. These dashboards make it possible to identify which student groups are most at risk and may benefit most from additional support structures like the Free Store.

Free Store Program Surveys

Finally, the most targeted data would come from surveys conducted through the Free Store itself. These surveys would be designed to measure students' experiences with the store, as well as indicators linked to persistence and enrollment decisions.

Questions might include:

- - "Access to the Free Store improved my sense of belonging and institutional pride as a UNCG student."
 - "I could not have remained enrolled if not for access to Free Store resources."

Surveys should also track frequency of use, types and value of items accessed, and perceived changes in stress, financial hardship, or sense of inclusion. Although this data would be self-reported, it would provide valuable insight into how students perceive the store's impact on their academic experience.

Analytic Possibilities

To evaluate the Free Store's potential impact, a difference-in-differences design could be utilized to track changes in student outcomes before and after the program's launch. This approach would help isolate the effect of the Free Store by accounting for broader trends and cohort-specific factors that might otherwise confound the results. With Free Store participation survey data available at the individual level, regression models could test associations between usage patterns and student outcomes such as retention, GPA, or credit accumulation, controlling for income, race, and first-generation status. If participation is anonymous, aggregate-level comparisons across demographic groups and time periods would still be possible.

Exploring Institutional Viability

By applying conservative, evidence-based assumptions throughout the data modeling process, we can be confident that a student-led, university-facilitated Free Store would provide substantial benefits to both the institution and its students. The data shows that a pilot project would attain net financial profitability for the university at .68% student utilization due to increases in tuition revenue. These gains are achieved exclusively through the well-researched enrollment and retainment benefits that result from additional financial aid. The Spartan Open Pantry, UNCG's traditional faith-based food aid program, served over 2,200 students in 2023 (Smith, 2024). Given this

existing level of need and engagement, it is reasonable to assume that a campus Free Store would reach the utilization levels required for institutional profitability.

Many intangible benefits were omitted from these models. Improvements to student workforce readiness, expansion of community-engaged curricula, reduction of stigma in accessing aid resources, and student empowerment gains, while difficult to quantify, are nonetheless extremely impactful in aggregate. Demonstrating that the Free Store is financially viable on its own, without relying on these additional benefits, further strengthens the case for piloting this program.

It is important to note that the model demonstrates diminishing benefit efficiency returns to scale. The assumptions underlying the pilot proposal – one university staff member, 800sq/ft of retail space, no corporate partnerships, lack of grant funding – inherently limit the ability of the Free Store to continue providing benefits efficiently beyond 3.38% utilization. At this threshold, additional staff, retail space, alternate inventory sourcing, and external funding should be explored. Expanding the program to locations within the local community could augment capacity and accessibility while serving an ambassadorial community-service role in the surrounding neighborhoods.

New Insights Gained from the Free Store Model

The model developed in this paper synthesizes insights from a range of disciplines, including economics, education, and community engagement, and translates them into a unified framework that can be expressed and evaluated in economic terms. While existing research does not fall short, it remains distributed across fields that rarely share a common modeling language. The Free Store model addresses that divide by operationalizing key assertions from the literature into a pilot-ready format that supports both theoretical insight and practical implementation.

The empirical relationship between financial aid and student persistence (Kane, 2003; Nguyen et al., 2019) provides a clear foundation for linking resource access to institutional revenue. Meanwhile, studies such as Idehai et al. (2024) emphasize the structural stigma that limits the use of traditional aid services, and Dresler and Tutt (2019) describe how mutual aid spaces support identity repair and belonging, often through a transition from recipient to contributor. The Free Store model brings these findings together by embedding reduced stigma, peer leadership, and participatory programming into a single structure. Internships and volunteer roles are not just operational necessities—they serve as opportunities for students to develop social capital, engage academically, and reshape how aid is experienced on campus. In this

way, the model remains both functional and measurable while also supporting long-term personal and institutional development.

This model also creates an opportunity to act on larger institutional goals. The UNCG Forward Together strategic plan and the Carnegie Classification for Community Engagement both identify civic learning, inclusion, and demonstrable student success as strategic priorities. The Free Store supports those goals while offering something often missing from community engagement initiatives: a mechanism for evaluating financial and academic returns. Unlike more diffuse programs, the model defines the roles of students, faculty, and institutional actors and ties them directly to outcomes.

Barr and McNamara (2022) and Coles–Ritchie et al. (2022) argue that student–designed and participatory programming improves relevance and sustainability. This model takes that principle seriously and extends it. By embedding participatory design within a structure that tracks retention and enrollment yield, it shows how community–based resources can function as institutional investments.

Most importantly, the model translates theory into praxis. It takes interdisciplinary insights spanning economics, identity, and engagement, and transforms them into a structure that can be tested, refined, and evaluated in practice. By modeling student usage, institutional return, and operational costs within a single system, the Free Store becomes more than a policy proposal. It becomes a living experiment in how values–driven programs can yield measurable economic outcomes. This reframing demonstrates how engagement and aid are not just moral imperatives, but strategic investments with tangible returns.

References

- Barr, M. L., & McNamara, J. (2022). Community-based participatory research: Partnering with college students to develop a tailored, wellness-focused intervention for university campuses. *International Journal of Environmental Research and Public Health*, 19(23), 16331. <https://doi.org/10.3390/ijerph192316331>
- Benjamin, D. J., Cooper, K., Heffetz, O., & Kimball, M. (2024). From happiness data to economic conclusions. *Annual Review of Economics*, 16(1), 359–391. <https://doi.org/10.1146/annurev-economics-081623-021136>
- Carnegie Classification of Institutions of Higher Education. (n.d.). *Community engagement elective classification*. <https://carnegieclassifications.acenet.edu/elective-classifications/community-engagement/>
- Coles-Ritchie, M., Power, C. A., Farrell, C., & Valerio, M. (2022). Pedagogy matters: A framework for critical community-engaged courses in higher education. *Gateways: International Journal of Community Research and Engagement*, 15(1). <https://doi.org/10.5130/ijcre.v15i1.8144>
- Dresler, E., & Tutt, H. (2019). Food insecurity, customer to volunteer transition and self-concept repair: A free store example. *Journal of Marketing Management*, 35(17–18), 1584–1599. <https://doi.org/10.1080/0267257X.2019.1683596>
- Idehai, O. V., Mbaya, P., Chung, T., & Bhurosy, T. (2024). A systematic review of factors associated with student use of campus food pantries: Implications for addressing barriers and facilitating use. *BMC Public Health*, 24(1), Article 17583. <https://doi.org/10.1186/s12889-023-17583-7>
- Janke, E. M., & Medlin, K. D. (2015). A centralized strategy to collect comprehensive institution-wide data from faculty and staff about community engagement and public service. *Metropolitan Universities*, 26(2), 125–146. <https://doi.org/10.18060/20981>
- Kane, T. J. (2003). A quasi-experimental estimate of the impact of financial aid on college-going (NBER Working Paper No. 9703). *National Bureau of Economic Research*. <https://doi.org/10.3386/w9703>
- Nguyen, T. D., Kramer, J. W., & Evans, B. J. (2019). The effects of grant aid on student persistence and degree attainment: A systematic review and meta-analysis of the

causal evidence. *Review of Educational Research*, 89(6), 831–874.
<https://doi.org/10.3102/0034654319877156>

Smith, R. (2024, January 6). UNCG expands on-campus food pantry. *WXII 12 News*.
<https://www.wxii12.com/article/university-of-north-carolina-greensboro-expands-food-pantry/46303050>

U.S. Department of Education. (2023). *Integrated Postsecondary Education Data System (IPEDS)*. <https://nces.ed.gov/ipeds/>

University of North Carolina at Greensboro. (2023). *Forward together: UNCG strategic plan*. <https://strategicplan.uncg.edu/forward-together/>

University of North Carolina at Greensboro. (2023). *Retention, graduation, and persistence dashboard*. <https://ire.uncg.edu/>